



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI JAYALAKSHMI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

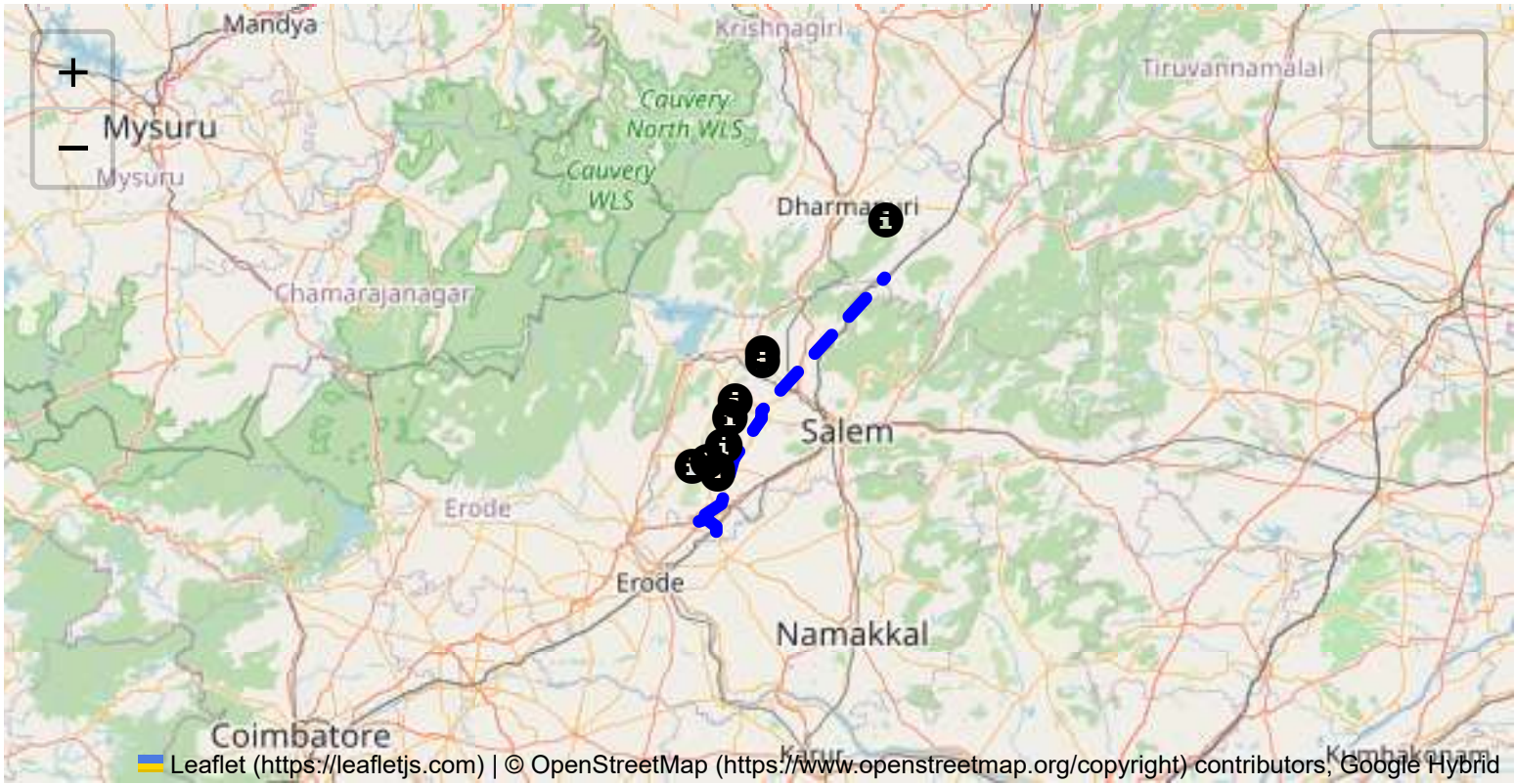
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

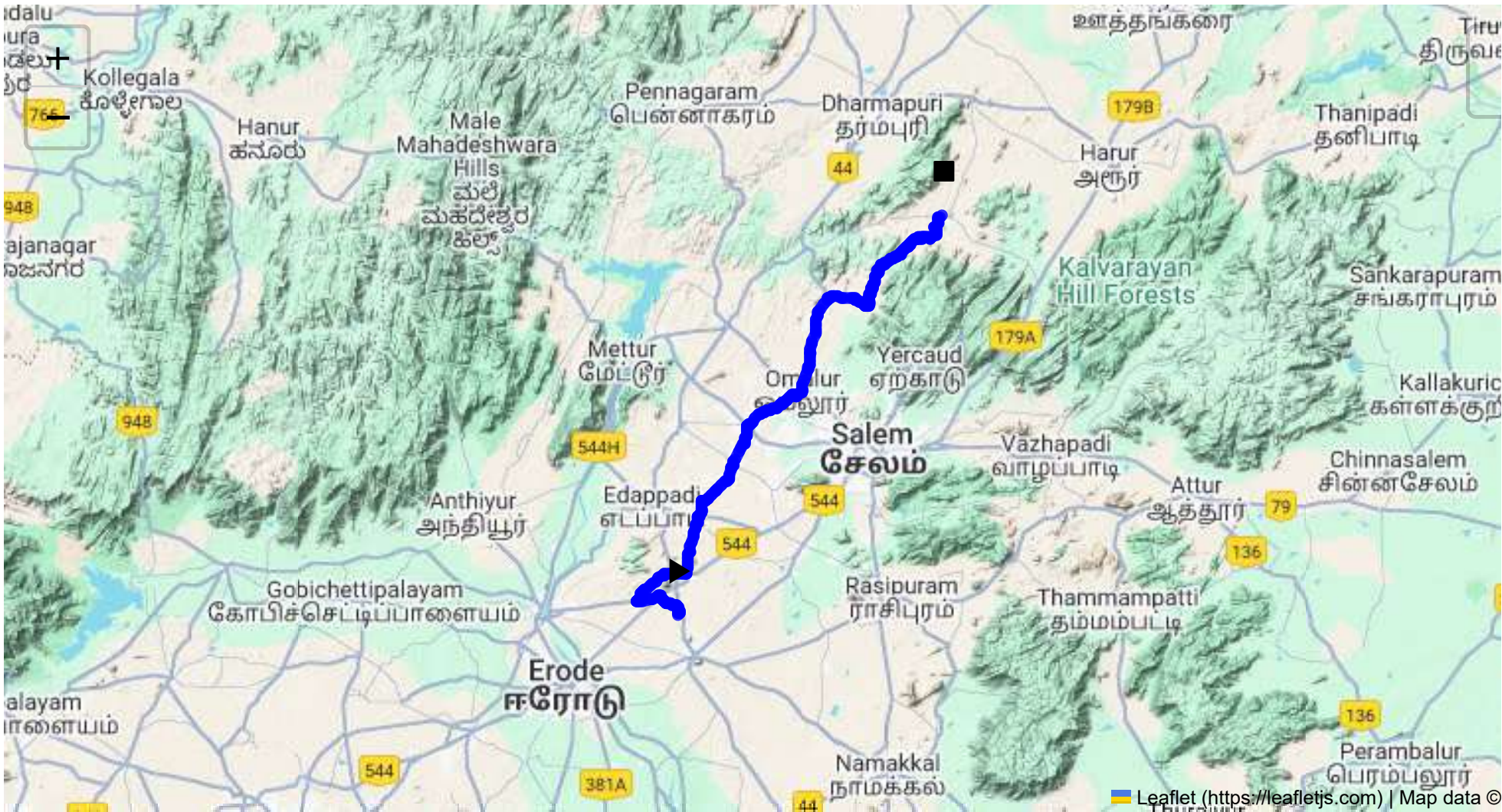
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 96.50 km
Estimated Duration: 2.3 hours
Adjusted Duration (Heavy Vehicle): 2.9 hours
Start: (11.4381, 77.8734)
End: (11.9817982, 78.2417402)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route extends from Sangagiri, Tamil Nadu, to Bommidi, Tamil Nadu, covering about 96.5 kilometers. It traverses various state highways and main roads like Sankari RS - Erode Privu Road, Bhavani Main Road, and SH 86. The journey includes small towns and rural areas with several connecting points important for navigation.

2. Typical Weather Conditions and Hazards: Tamil Nadu experiences a tropical climate with hot summers, monsoon seasons, and mild winters. The monsoon period, from June to September and again in October to December, can cause heavy rains leading to waterlogging and slippery roads. Wind conditions are generally moderate but can escalate during monsoon storms, posing risks of reduced visibility and control.

3. Traffic Patterns and Congestion: Traffic peaks during local rush hours (7:00-10:00 AM and 5:00-8:00 PM) in urbanized pockets like Sankari and Tharamangalam. Areas near market hubs and intersections like the Bhavani Main Road are prone to congestion. Rural stretches may not have significant traffic but be alert for unexpected slow-moving farm vehicles.

4. Road Quality and Infrastructure: The quality of roads varies along the route. State highways like SH 86 are generally well-maintained, but smaller roads may have potholes or uneven surfaces. Be cautious in less-illuminated rural areas at night. Infrastructure includes occasional pedestrian crossings and minor bridges.

5. Alternative Routes for Emergencies: For emergencies, diversions via larger highways (like NH 47 or NH 544) should be considered for less populated routes offering faster aid. However, these alternatives may add considerable distance, so they should primarily be used for major disruptions.

6. Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials is regulated under local guidelines, requiring permits. Nighttime travel might be restricted in certain areas to minimize risks. Ensure compliance with Tamil Nadu's specific transport policies for hazardous materials.

7. Historical Incidents Involving Heavy Vehicles: Some isolated incidents have been noted involving heavy vehicles, mainly due to brake failure on downhill stretches or collisions caused by narrow passes. No major incidents of hazardous materials spills have been recorded, but caution is advised while transporting.

8. Environmental Considerations: Be aware of nearby agricultural areas that may be sensitive to chemical spills. Preserve natural surroundings by minimizing exposure to sensitive regions and adhering to designated routes. Local wildlife crossings, though rare, should be anticipated.

9. Communication Coverage: Most areas along this route have stable cellular coverage, especially near towns. Nevertheless, occasional dead zones may occur near rural stretches and hillside paths like those in Koranampatti or Ayveli.

10. Estimated Emergency Response Times: Emergency services are typically accessible within 30-45 minutes in towns but may extend to over an hour in remote areas. Comprehensive emergency services are available in nearby towns such as Salem for serious incidents.

11. Overall Summary of Risk Assessment: This route presents moderate risk primarily due to varied road conditions and intense weather during monsoon seasons. Light traffic in rural areas reduces collision risks, but vigilance is required near urban centers. The regulatory landscape demands awareness from drivers regarding transport permits and safe driving standards for hazardous materials. Proactively planning for alternative routes and understanding emergency protocols is essential to safely transporting hazardous goods along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57035, 77.90223	30 KM/Hr
13	Turn	High	11.57041, 77.90221	15 KM/Hr
14	Turn	Medium	11.57649, 77.90610	30 KM/Hr
15	Turn	Medium	11.58279, 77.90658	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	High	11.69595, 77.97114	15 KM/Hr
21	Turn	Medium	11.71386, 77.99552	30 KM/Hr
22	Turn	Medium	11.71735, 78.00019	30 KM/Hr
23	Turn	Medium	11.72765, 78.02291	30 KM/Hr
24	Turn	Medium	11.73786, 78.03505	30 KM/Hr
25	Turn	High	11.73652, 78.03866	15 KM/Hr
26	Turn	Medium	11.74110, 78.04575	30 KM/Hr
27	Turn	Medium	11.74257, 78.04708	30 KM/Hr
28	Turn	Medium	11.77998, 78.05786	30 KM/Hr
29	Turn	Medium	11.86915, 78.08382	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	High	11.87099, 78.08358	15 KM/Hr
31	Turn	High	11.87110, 78.08383	15 KM/Hr
32	Turn	Medium	11.87072, 78.08439	30 KM/Hr
33	Turn	Medium	11.87088, 78.08556	30 KM/Hr
34	Turn	High	11.87246, 78.08704	15 KM/Hr
35	Turn	Medium	11.86898, 78.12074	30 KM/Hr
36	Turn	Medium	11.86495, 78.12530	30 KM/Hr
37	Turn	High	11.85820, 78.13580	15 KM/Hr
38	Turn	Medium	11.86044, 78.13937	30 KM/Hr
39	Turn	Medium	11.86652, 78.13879	30 KM/Hr
40	Turn	Medium	11.88411, 78.14478	30 KM/Hr
41	Turn	Medium	11.88540, 78.14474	30 KM/Hr
42	Turn	Medium	11.89899, 78.14930	30 KM/Hr
43	Turn	Medium	11.90581, 78.15292	30 KM/Hr
44	Turn	High	11.90760, 78.15205	15 KM/Hr
45	Turn	Medium	11.91226, 78.15594	30 KM/Hr
46	Turn	Medium	11.91312, 78.15616	30 KM/Hr
47	Turn	Medium	11.91519, 78.15837	30 KM/Hr
48	Turn	High	11.92454, 78.17073	15 KM/Hr
49	Turn	Medium	11.92448, 78.17340	30 KM/Hr
50	Turn	Medium	11.92820, 78.18040	30 KM/Hr
51	Turn	Medium	11.92797, 78.18165	30 KM/Hr
52	Turn	Medium	11.93585, 78.18730	30 KM/Hr
53	Turn	Medium	11.95116, 78.20647	30 KM/Hr
54	Turn	Medium	11.95200, 78.20703	30 KM/Hr
55	Turn	High	11.95258, 78.20848	15 KM/Hr
56	Turn	Medium	11.95173, 78.21017	30 KM/Hr
57	Turn	High	11.95221, 78.21173	15 KM/Hr
58	Turn	Medium	11.95165, 78.21276	30 KM/Hr
59	Turn	Medium	11.95228, 78.21413	30 KM/Hr
60	Turn	High	11.95220, 78.21522	15 KM/Hr
61	Turn	Medium	11.95288, 78.21603	30 KM/Hr
62	Turn	Medium	11.95264, 78.22467	30 KM/Hr
63	Turn	High	11.95178, 78.22600	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
64	Turn	Medium	11.95847, 78.23410	30 KM/Hr
65	Turn	Medium	11.96509, 78.23292	30 KM/Hr
66	Turn	High	11.96652, 78.23302	15 KM/Hr
67	Blind Spot	Blind Spot	11.96679, 78.23211	10 KM/Hr
68	Turn	High	11.97329, 78.23713	15 KM/Hr
69	Turn	High	11.97765, 78.23556	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
12	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
27	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
31	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
32	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
33	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
34	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
35	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
26	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57035, 77.90223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57041, 77.90221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58279, 77.90658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69595, 77.97114



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71735, 78.00019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72765, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73786, 78.03505



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74110, 78.04575



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87099, 78.08358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87110, 78.08383



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87072, 78.08439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87088, 78.08556



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87246, 78.08704



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86898, 78.12074



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95116, 78.20647



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97329, 78.23713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97765, 78.23556

